FIT3161 Computer Science Project 1

Final Proposal Report

Privacy-Preserving AI for Detecting Deepfake Profile Photos

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1.0 Introduction

1.1 Project Overview

The MCS24 Team is dedicated to develop *FaceGuard: a Privacy-Preserving AI for Detecting Deepfake Profile Photos*, an innovative technology created to identify and classify AI-generated human face images on social and dating platforms. By integrating powerful deep learning algorithms with explainable AI, the technology improves online identity security through transparent and responsible deepfake detection. Our team is made up of four dedicated members, comprising *Er Jun Yet*, *Tan Yan Yi*, *Bryan Ong* and *Ali Afzal* under *Dr. Bisan A N Alsalibi*'s supervision, applies Agile principles to deliver an accurate and trustworthy AI-based software solution that fosters digital trust and safety.

1.1.1 Project Context

In today's digital world, the rapid growth of generative AI has made it more challenging for people to differentiate between real and fake online identities. Social engineering frauds, catfishing, and misinformation frequently take use of these synthetic media, diminishing the trust on social and dating platforms. The *FaceGuard* project supports global efforts in responsible AI governance and privacy-preserving technologies, highlighting the necessity of explainability, transparency and ethical AI deployment in social media ecosystems

1.1.2 Problem Statement

The rising misuse of AI-generated face images in online profiles poses a serious threat to user safety, privacy and digital trust.

1.1.3 Project Expectation

FaceGuard is anticipated to deliver a fully functional web-based system with visual explainability via *Grad-CAM* heatmaps that can accurately classify profile photos as real or AI-generated. In addition, the project also involve dataset collection, model training with *CNN* architectures and a user-friendly interface hosted using *Netlify*. Beyond technical results, the system intends to raise public awareness of AI abuse and create a reproducible framework for ethical deepfake detection.

1.2 Report Structure

Section 1	Overview of project introduction, context, problem statement and expected outcomes
Section 2	Literature review on deepfake detection, explainability, ethics and project rationale
Section 3	Project management, involving scope, organisation, schedule and resource planning
Section 4	Proposed methodology, outlining data flow, model implementation and development process
Section 5	Conclusion and summary of project

2.0 Literature Review

2.1 Project Background

The advancement of *GenAI* [1], especially *GANs* [2] and *DMs* [3], has led to the widespread of realistic AI-generated human faces in the online world, which despite advancing digital creativity, but also being exploited for deepfakes for identity theft, impersonation and even romance scams, thereby eroding public trust and also complicating the verification of digital authenticity [9][35]. Several deepfake detection models, like convolutional and transformer-based architectures, such as *Xception* [4] and *ViTs* [5], were invented with strong accuracy in benchmark datasets; however, practical deployment remains limited by concerns of fairness in demographics, transparency in generative techniques and privacy in user data [33][17]. In essence, existing online detection tools seldom explain their decision process or safeguard uploaded user data, constructing a "black-box" system that risks bias and privacy violations [30][21].

2.2 Literature Findings

2.2.1 Threat Landscape and Manipulation Taxonomy

The democratisation of open-source frameworks, such as *StyleGAN*, *DeepFaceLab* and *Stable Diffusion*, has reshaped the entire digital threat landscape for individuals with no programming knowledge to generate convincing fake media of synthetic human faces from real photographs [12][19][25]. Current research has discovered four key principal categories in facial manipulation, which are identity-swap, attribute manipulation, expression or lip-sync manipulation and complete synthetic generation [33], as detailed in Appendix 7.1.1 *Taxonomy of Deepfake Manipulation*. Each category exhibits unique forensic artefacts and varying compression patterns, ranging from local pixel discrepancies to global illumination anomalies, complicating robust detection in AI-generated images [17]. This new emergence of cybersecurity threats and ethical risks extends beyond individual deception but to broader societal disruption [9][24], as detailed in Appendix 7.1.2 *Taxonomy of Deepfake Threats*.

[1] GenAI - *Generative AI* (GenAI)
[2] DMs - *Diffusion Models (DMs)*
[3] GANs - *Generative Adversarial Networks* (GANs)
[4] Xception - *Extreme Inception* (Xception)
[5] ViTs - *Vision Transformers* (ViTs)

2.2.2 Detection Datasets and Benchmarks

The success of any deepfake detection system solely depends on the availability of comprehensive and diverse datasets that include clear visuals, diverse demographics and facial manipulation. In the early years, notable libraries like *FaceForensics++*, *DeepFake Detection Challenge* and *Celeb-DF* played a pivotal role in providing a significantly large volume of real and manipulated human face collections for model training [26][7][15]. However, models trained on these datasets achieved strong in-sample accuracy but obtained poor generalisation and unknown demographic bias when new datasets applied, because most datasets were video-level benchmarks, which are inapplicable for AI image detection from *GANs* and *DMs*, with the inherent limitation of demographic diversity and minimal documentation of fairness attributes [16][4], as detailed in Appendix 7.1.3 *Analysis of Existing Major Deepfake Datasets*. As the result of the insufficient use for studying static image authenticity and fairness analysis, a newer dataset like *AI-Face* introduced image-level and fairness-aware benchmarks with one of the largest and most demographically balanced repositories of real and AI human faces from *DMs*, with even metadata annotations on gender, age and skin tone. With such high granularity in dataset, *AI-Face* is an ideal choice for our *FaceGuard* project in achieving a balanced and equitable model evaluation.

2.2.3 Detection and Explainability Approaches

Deepfake detection research has evolved extensively, from convolutional models of early *CNNs* such as *XceptionNet* and *MesoNet* that focused on pixel-level inconsistencies with strong in-sample accuracy only, as well as *ResNet* and *EfficientNet* that introduced lightweight attention modules with deeper hierarchical spatial-frequency features yet still lacked detection of subtle global patterns by diffusion-model manipulation [14], to resource-intensive *ViTs* that employed holistic self-attention modules to capture long-range dependencies with strong generalisation across any manipulation sources [6][5]. Then later, a hybrid *CNN-ViT* framework is invented to combine convolutional feature extractors with transformer encoders to equip both local-texture precision and global-context reasoning, achieving the art of stability in robustness and fairness, but introducing greater architectural complexity and training overhead [31]. A thorough analysis of architecture alternatives is detailed in Appendix 7.1.5. Moreover, deepfake detection research also takes into account in algorithmic transparency and explainability approaches to convert hidden model inferences into interpretable evidence for users and developers. Common *XAI* [6] integrations have been widely used, like *Grad-CAM* [7] that produces intuitive visual

[6] XAI - *Explainable AI* (XAI)
[7] Grad-CAM - *Gradient-weighted Class Activation Mapping* (Grad-CAM)

saliency map with coarse spatial localisation at low computation cost, *LIME* that offers model-agnostic design that provides case-specific reasoning for developer audits and fairness checks, and more advanced alternative such as *SHAP* that offers game-theoretic feature attribution to quantify each input's contribution to a result for bias detection and consistency analysis [22][18][23]. A thorough analysis of explainability alternatives is detailed in Appendix 7.1.4.

2.2.4 Fairness, Ethics, Privacy and Governance

Studies have found that most face-analysis models exhibit demographic bias with uneven performance across gender and skin-tone groups, suggesting that technical accuracy alone is insufficient [3][24]. For instance, several studies have discovered that most face-analysis algorithms display systemic demographic bias, performing less accurately on darker-skinned and female faces due to skewed data representation [3]. Parallel to fairness is the importance of privacy preservation, because many online detection platforms store user-uploaded images on central servers for future analysis, causing violations of data-protection laws in *GDPR* and *PDPA 2010* [32][11]. Therefore, user privacy should be protected through ephemeral, in-memory inference, encrypted transmission and metadata removal, consistent with the *NIST*'s privacy-by-design principles [20]. At a governance level, all AI projects still must align with the ACM Code of Ethics [1] and the EU Ethics Guidelines for Trustworthy AI [8] to practice accountability, human control and societal benefit [10]. As such, transparency must be reinforced through *XAI* techniques, demystifying the reasoning of adopted AI models and building informed trust [30].

2.2.5 Project Management Process

Several process models have been considered for our project development. Traditional model like Waterfall Model follows a strict sequential flow of requirement, design, development, testing and deployment phases, which is inflexible for deep learning projects like *FaceGuard*, where model accuracy, dataset training and fairness metrics evolve dynamically [27]. Conversely, the Spiral Model combines iterative prototyping with flexible and structured risk assessment, which is inapplicable for its heavy documentation load and cost-estimation requirements for a university-level project like *FaceGuard* [2]. Finally, the Scrum Model emphasises short development cycles called sprints with adaptive planning and consistent stakeholder collaboration, which is perfect for *FaceGuard* with multiple objectives to be achieved by sprints to deliver partial prototypes to the final MVP [28]. After a critical analysis as detailed in Appendix 7.1.6, by adopting Agile Scrum Model, *FaceGuard* project benefits from flexibility in delivering a holistic AI detection platform that remains adaptable to stakeholder feedback and emerging research standards.

2.3 Project Design and Rationale

In conclusion, the preceding literature review highlights many existing deepfake detection models achieving impressive accuracy yet frequently lacking fairness evaluation, AI explainability and privacy-preserving deployment. Now, the design of *FaceGuard* was therefore conceptualised to bridge these research gaps by adopting a three-tier architecture, comprising an AI detection engine, a user-centric web interface and a secure backend system. At the AI layer, *FaceGuard*'s initial MVP employs a *ResNet*-based model developed in *PyTorch* to detect real and AI-generated human faces for its stable performance, moderate computational demand and full compatibility with *Grad-CAM* explainability feature. Once the MVP achieves baseline accuracy and stability, *FaceGuard* will potentially be further developed into a hybrid *CNN-ViT* model to achieve both local-texture precision and global-context reasoning, enhancing cross-dataset generalisation and fairness evaluation with the large-scale diversity of demographics in *AI-Face* dataset. Solely by this reason, Agile Scrum Model is adopted to develop *FaceGuard* efficiently in MCS24 Team. At the web layer, a simple dashboard with *React.js* is built with an upload feature, a result interface and an AI visualisation heatmap and its confidence score. A lightweight *Grad-CAM* visualisation is optimal enough for real-time inference on uploaded images to promote transparency to users. At the security layer, the web system is developed with *FastAPI* that handles asynchronous request handling for efficient concurrent users and supports HTTPS encryption with JSON-based API responses [29]. *FaceGuard* decided to implement temporary image processing in volatile memory, where images are automatically deleted after used and EXIF metadata of images are stripped before inference, where no user data is stored in any ways, aligning to our privacy-by-design requirement. As justified in our Appendix 7.1.7 *Gap Analysis of Existing Deepfake Models and FaceGuard Design* and analysed in our Section 2.2 *Literature Findings*, *FaceGuard* is determined in the dual commitment of both technical reliability and ethical integrity.

3.0 Project Management

3.1 Project Scope

3.1.1 In-Scope

- **AI Classifier Model:** A deepfake image detection model using the *AI-Face* dataset and fine-tuned *ResNet* architecture for MVP.
- **Explainability Module:** A *Grad-CAM* visualisation saliency map that visually justify the reasoning behind real and AI image classification.
- **Web Application Interface:** A *React*-based dashboard for drag-and-drop image uploads, classification results, confidence scores and explanatory heatmaps.
- **Privacy-preserving Mechanism:** Temporary in-memory inference, auto-deletion of user uploads and metadata removal..

3.1.2 Out-of-Scope

- Facial recognition, identity matching or biometric comparison.
- Mobile or native-app deployment.
- Video deepfake or multimodal content detection.
- Integration with external commercial APIs beyond open-source datasets and frameworks.

3.1.3 Constraints

- **Resource Constraints:** Limited GPU availability, reliance on cloud services such as *Google Colab Pro+* and Monash GPU lab.
- **Time Constraints:** Limited to the university semester schedule of two semesters.
- **Ethical Constraints:** Compliance with the ACM Code of Ethics and strict adherence to privacy-by-design.

3.1.4 Assumptions

- *AI-Face* dataset is representative of current deepfake outcomes from *GANs* and *DMs*.
- Users upload static images containing a clearly visible human face.
- Project stakeholders (Supervisor and Teaching Team) will provide regular sprint feedback.
- All open-source libraries used remain actively maintained throughout the development.

3.2 Project Requirements

3.2.1 Product Requirements Consultation

The following approaches were carried out by the MCS24 Team in the requirement elicitation phase.

- **Supervisor Consultation:** The MCS24 Team engaged in regular consultations with the project supervisor, *Dr Bisan A. N. Alsalibi* to establish the foundational knowledge and specifications of the *FaceGuard* project, providing a high-level overview of the initial requirements.
- **Project Specification Review:** The MCS24 Team reviewed and analysed the detailed project specification documents and relevant research papers of the *FaceGuard* project, solidifying our understanding to the project.
- **Mockup Interface Design:** The MCS24 Team designed a draft mockup interface of the *FaceGuard* web platform, providing a comprehensive idea of the FRs [8] and NFRs [9].
- **Requirement Traceability Matrix (RTM):** The MCS24 Team submitted a RTM documentation with a detailed list of confirmed FRs and NFRs to be finalised and approved by *Dr Bisan* prior to the actual project execution.

3.2.2 Requirement Traceability Matrix (RTM)

The detailed list of final requirements of the *FaceGuard* project has been finalised in Appendix 7.2.

3.2.3 Product Acceptance Criteria

The detailed list of user stories along with its acceptance criteria of the *FaceGuard* project has also been prepared in Appendix 7.3.

[8] Functional Requirements (FR): Essential functions or behaviors that the system needs to have.
[9] Non-Functional Requirements (NFR): security, usability, scalability, and ethical compliance being examples of quality attributes.

3.3 Project Organisation

3.3.1 Process Model

The MCS24 Team adopted the **Agile Scrum methodology** to manage the *FaceGuard* project effectively and flexibly, after considerations detailed in Section 2.2.5. The MCS24 team operates on **two-week sprint cycles**, each comprising of Sprint Planning, Sprint Development, Sprint Review and Sprint Retrospective. Prior to **Sprint Planning**, the Product Owner prioritises user stories from the **Product Backlog** based on the project value and feasibility as well as stakeholder interests, obtaining a clear sprint goal that aligns with *FaceGuard*'s objectives of a privacy-preserving and explainable detection model. During the Sprint Planing, the **Sprint Backlog** defines detailed tasks assigned to all members along with the acceptance criteria. The development of *FaceGuard* follows continuous integration, where developers collaborate across backend, AI and UI components to deliver a functional MVP at every sprint. Regularly, weekly **Stand-up meetings** is held for weekly updates, while **Sprint Reviews** and **Sprint Retrospectives** provide opportunities for feedback, reflection and improvement in both project development and management. This iterative process as shown in Figure 3.1, allows *FaceGuard* to adapt efficiently to evolving technical and ethical requirements, ensuring consistent alignment with the project's goal and stakeholder requirements of delivering a transparent and responsible AI detection system.

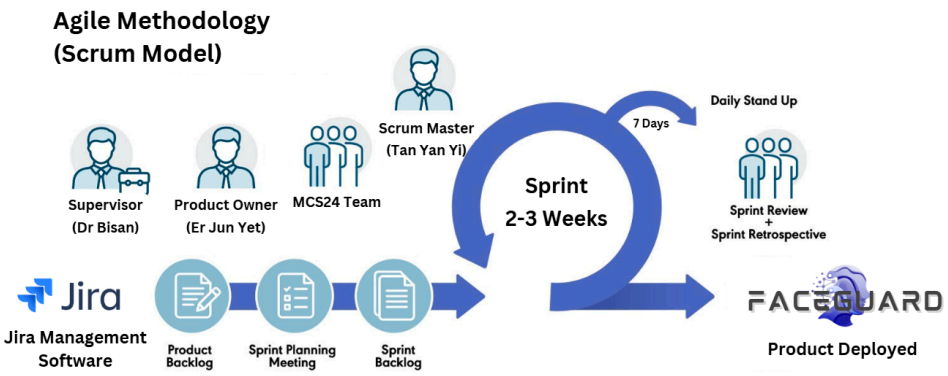

Figure 3.1: *FaceGuard* Development in Agile Scrum Model

3.3.2 Team Organisation

The full organisation structure of the MCS24 Team is presented in Appendix 7.5, with the adoption of Agile methodology as introduced in Section 3.3.1 and the RACI Matrix [10] as detailed in Appendix 7.4.

3.4 Project Management

3.4.1 Product Monitoring System

The MCS24 Team employs *Jira* as the primary Product Monitoring System to effectively track project progress and maintain alignment with milestones, with the following key components.

- **Product Backlog System:** A comprehensive repository of all project requirements and user stories for Product Owner, to prioritise tasks according to sprint goal.
- **Sprint Backlog System:** A subset of selected tasks from the Product Backlog by Product Owner (PO) and stakeholders for the current two-week sprint, to provide a clear, actionable view of current team commitments.
- **Kanban Board:** A board of tasks placed in columns of To-Do, In-Progress and Done, to quickly identify workflow bottlenecks and ensure timely task completion.
- **Burndown Chart:** A graph of the remaining workload against the sprint timeline, to act as a real-time tracker to meet our sprint goals and milestones.
- **Sprint Review and Retrospective:** A meeting with all members and stakeholders to present project deliverables for supervisor feedback and to reflect on team performance and improvements for the upcoming sprints.

3.4.2 Communication System Policy

The MCS24 Team also adopts the Agile Scrum ceremonies such as Standup Meeting, Sprint Planning, Sprint Retrospective, Sprint Review and Emergency Meeting, into the official Communication System Policy as detailed in Appendix 7.6.

3.4.3 Version Control Policy

The MCS24 Team utilises *GitHub* as the primary version control for smooth and traceable collaboration with the *FaceGuard* repository to track clear history of the project development, hence the Version Control Policy is divided into two key components.

Branching Policy:

- `master` branch serve as the stable production-ready branch that has undergo strict review and testing by the QA/QC Specialist and the PO.

- `developer` branches are the workspace branch for respective developers to develop their assigned feature.

Commit Policy:

- Done code must be checked and confirmed by the QA/QC and PO before merging.
- Commit messages must be descriptive and meaningful with a clear explanation.
- Regular pulling and pushing of code should be practised prior and upon completion of task.

3.4.4 Quality Control Policy

The QA/QC Specialist is responsible for regulating the Quality Control Policy and assuring the code and product quality.

- All tests must created by the QA/QC to prevent bugs and potential issues.
- All the code changes must be tested by QA/QC and approved by PO prior to merging.
- A basic code review checklist is prepared in Appendix 7.7 as a reference for a good coding standard.

3.4.5 Risk Management System

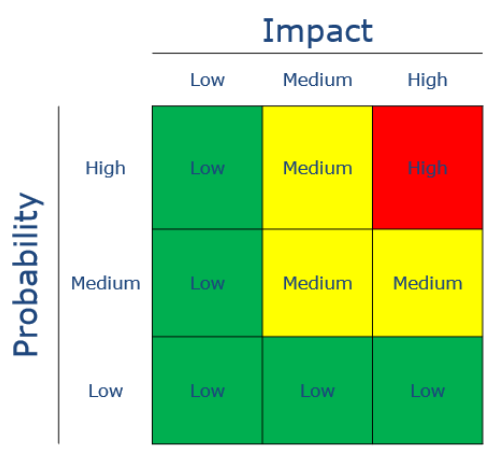

Figure 3.2: Probability/ Impact Matrix

The project adopts the **SWOT analysis** as a tool to categorise the risks into internal and external factors as shown in Appendix 7.9. To analyse the risks, we will use a probability/impact matrix, as shown in Figure 3.2, to classify the risks based on the probability of these risks occurring and the potential impact of the risks if it were to happen. This classification allows for us to focus on the risks that have a higher severity to ensure they do not affect the team's progress in the future. We also employ the **TARA framework** to develop strategies for each of the risks as shown in Appendix 7.10. The **Risk Register**, as detailed in Appendix 7.8, ensures that risks are continuously tracked throughout the whole project.

3.4.6 Documentation System

The MCS24 Team has a documentation system to ensure that all the requirements and references in the project are documented well in *Google Drive*. This includes meeting minutes, project requirements, Gantt Chart, RTM, WBS and other documentation.

3.5 Project Schedule

The *FaceGuard* project's schedule is made to align with the Agile methodology adopted by our team, which emphasises iterative development, continuous feedback and flexible planning. The project is divided into three major phases which are the **Research**, **Development** and **Findings** phases over the course of two academic semesters. For the **first semester**, we will be focusing mainly on research, literature review, dataset acquisition, and data preprocessing. For the **second semester**, we will be focusing mainly on model training, system integration testing and final deliverables.

3.5.1 Standard Commitment Hours

Each team member contributes roughly 21 to 23 hours per sprint throughout the project lifecycle, including development and testing phases. This adds up to an estimated 89 total man-days as detailed in Appendix 7.13, which guarantees constant progress in all phases in accordance with Agile iteration cycles and milestone goals.

3.5.2 Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS), given in Appendix 7.11, outlines the FaceGuard project using a hierarchical organisation of phases and tasks to ensure an effective project execution.

3.5.3 Gantt Chart

The Gantt Chart, given in Appendix 7.12, illustrates the entire project timeline, outlining task sequences, estimated durations, dependencies and milestone targets.

3.6 Resource Requirements

3.6.1 Human Resource

The MCS24 Team comprises four committed members with specialised roles according to their area of expertise. All members collaborate together with the adoption of Agile Scrum methodology with the team structure as detailed in Section 3.3.2, supervised by *Dr. Bisan*.

3.6.2 Technical Resource

Table 3.1: Software Specifications

Software	Specification
Operating System	*Windows 10* and above
Programming Language	*Python 3.14*
IDE	*Visual Studio Code 1.105.1*
Database System	*MongoDB 8.0.15*
Deep Learning Framework	*PyTorch 2.8.0*
Inference Engine	*ResNet / CNN-ViT*
Explainability Service	*Grad-CAM*
Backend Web Framework	*FastAPI 0.120*
Frontend Web Framework	*React 19.2*
Web Application Hosting	*Netlify*
Image Metadata Removal Tool	*ExifTool 13.40*

Table 3.2: Hardware Specifications

Hardware	Specification
Processing	*Intel Core i7* or *Apple M2* chip (multi-threaded AI processing)
Memory	Minimum 16 GB memory to support concurrent development
Storage	512 GB NVMs SSD for faster file access
Networking	Reliable internet connection using Wi-Fi 6
Alternative	*Google Colab Pro +*

4.0 Project Methodology

4.1 Software and Tools

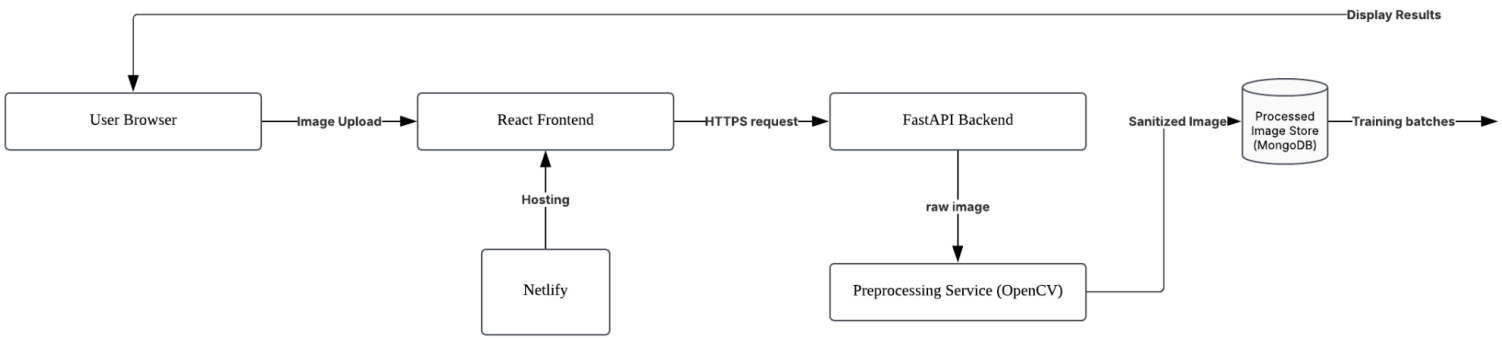

Figure 4.1: System Architecture Diagram (Part 1)

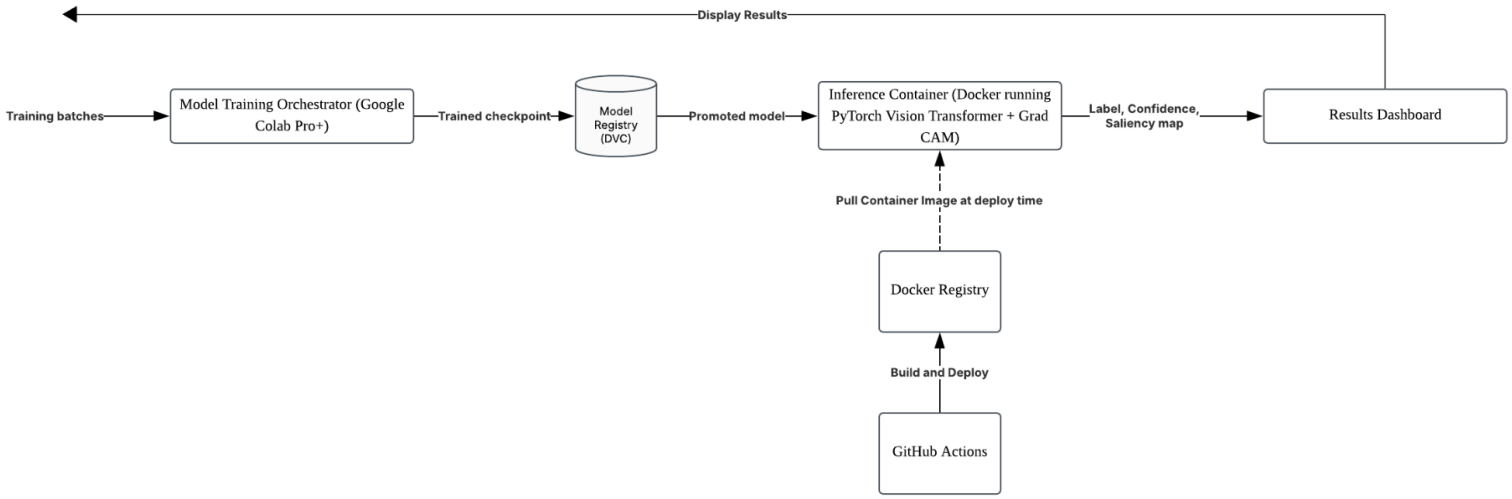

Figure 4.2: System Architecture Diagram (Part 2)

The overall structure and interaction of these software and tools are shown in the System Architecture Diagram (Figure 4.1 & 4.2).

The *FaceGuard* system is implemented with a specified software for each role: *React* provides the user interface, *FastAPI* hosts the backend, *OpenCV* undergoes metadata removal and normalization, *PyTorch* starts *ViT* inference, Grad-CAM gives explainability visualisations, *MongoDB* stores processed images, *DVC* tracks dataset, Docker packages the runtime environment, *GitHub* Actions automates CI/CD, *Google Colab Pro+* provides GPU capacity, and Netlify hosts the static frontend.

4.2 Step-by-Step Methodology

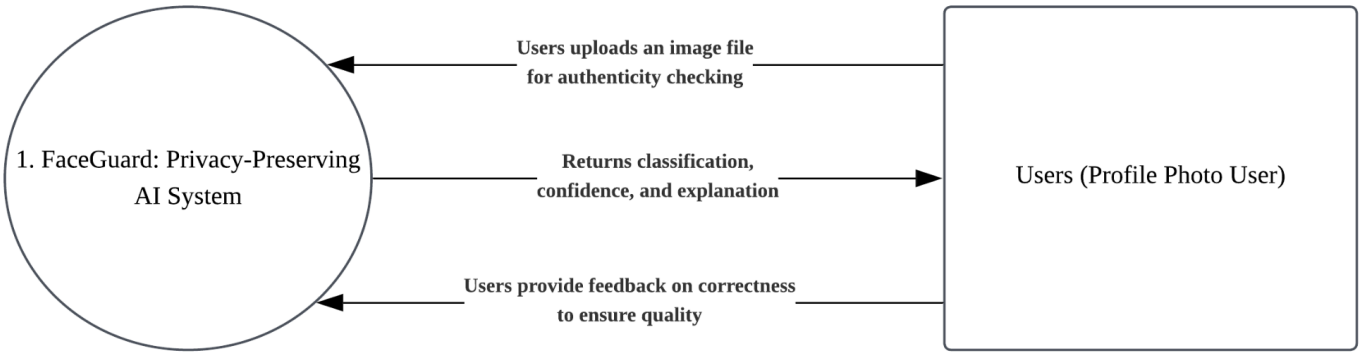

Figure 4.3: Level 0 Data Flow Diagram

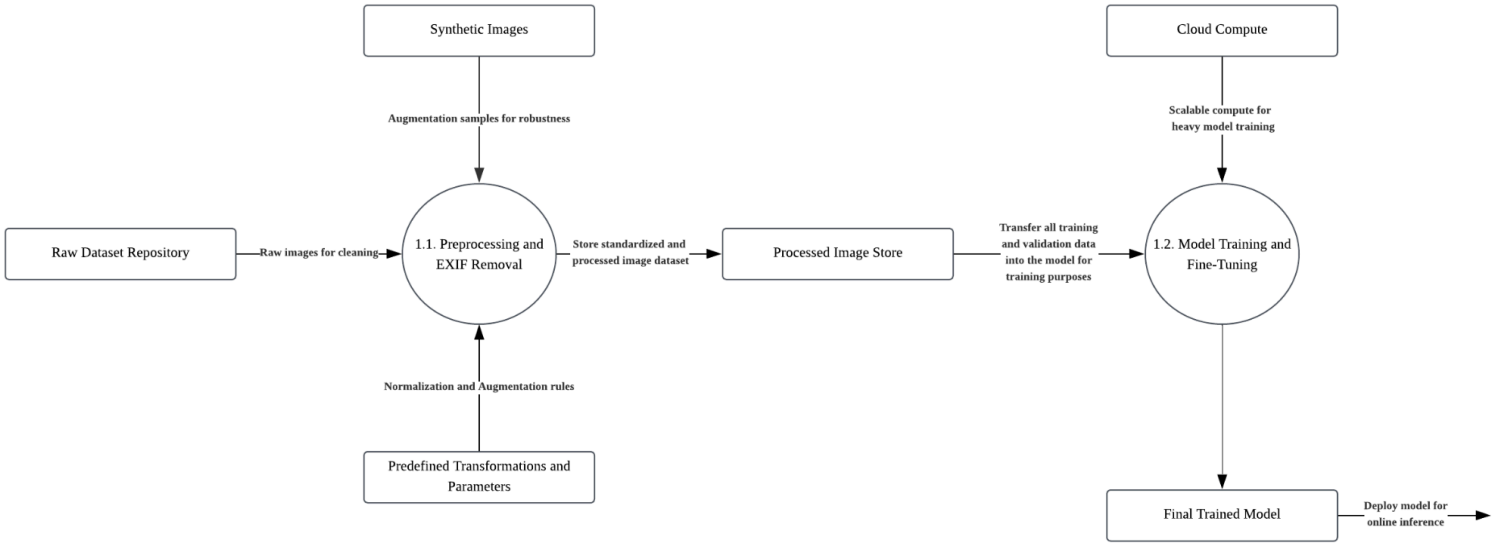

Figure 4.4: Level 1 Data Flow Diagram (Part 1)

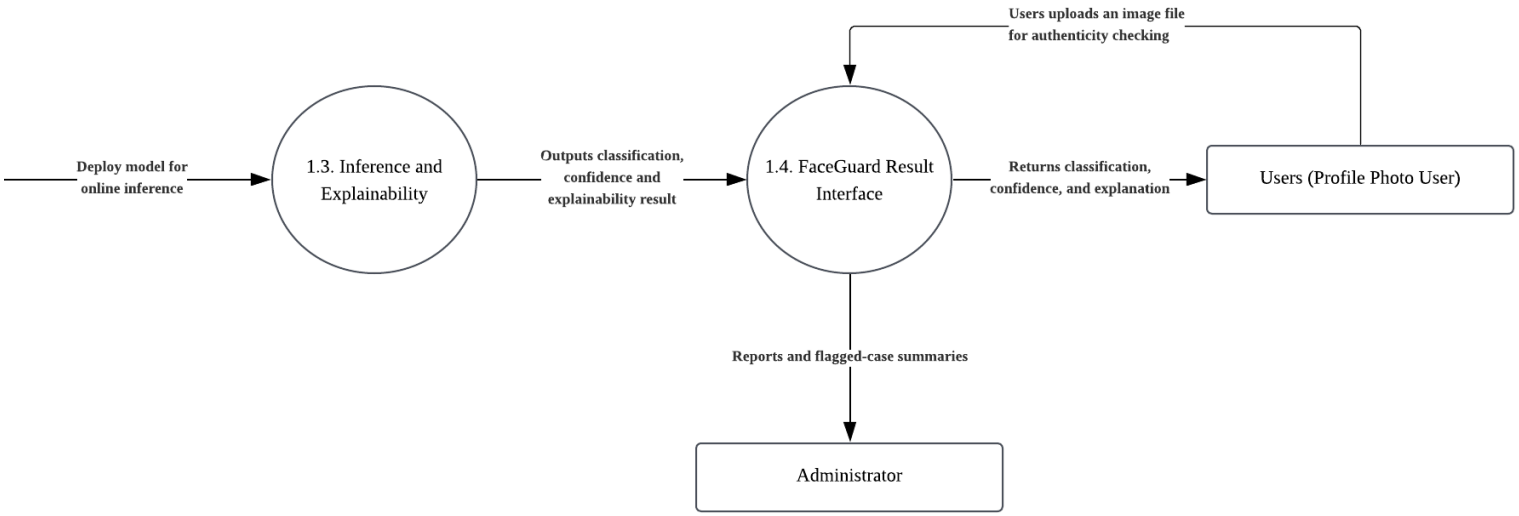

Figure 4.5: Level 1 Data Flow Diagram (Part 2)

The data flow of the entire *FaceGuard* system is first shown in the Level 0 Data Flow Diagram (Figure 4.3) and elaborated more in the Level 1 Data Flow Diagram (Figure 4.4 & 4.5).

Data flows through the system in a deterministic sequence, *React* frontend receives images then posts them to *FastAPI*; The backend sends raw images to the *OpenCV* preprocessing service for metadata removal; *MongoDB* stores the sanitized images; The promoted model checkpoint is loaded into the Dockerized *PyTorch* inference container; Each inference will generates a saliency map using *Grad-CAM*; Finally, the classification, confidence and saliency outputs are returned to the dashboard.

4.3 Justification and Reproducibility

The reproducible design for *FaceGuard* system is supported by the modular structure shown in the System Architecture Diagram (Figure 4.1 & 4.2).

Every software and tools are selected based on its reproducibility, performance and scalability. *DVC* makes sure data and model lineage, *Docker* ensures consistent execution across environments, *PyTorch* and the *ViT* offer reliable model portability, and *Github* Actions facilitates reproducible deployment. To guarantee traceable development, every experiment will record dataset versions, commit hashes, and hyperparameters.

5.0 Conclusion

The MCS24 Team's dedication to creating a safe, open, and moral AI system for identifying deepfake profile pictures is demonstrated by the *FaceGuard* project. The proposal, which is based on explainable and privacy preserving AI principles, has a clear scope, proven requirements, and an Agile management plan that guarantees stakeholder alignment and iterative development.

FaceGuard aims to provide a reliable solution that improves online safety while maintaining accountability by combining deep learning, *Grad-CAM* explainability, and secure web deployment. Technical feasibility and disciplined project execution are demonstrated by the organised approach, which is backed by resource allocation, risk management, and acceptance criteria.

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7.0 Appendix

7.1 Additional Literature

7.1.1 Taxonomy of Deepfake Manipulation

Table 7.1: Taxonomy of Facial Manipulation Techniques [19]

Category	Description	Tools / Models	Misusage	Challenge
Identity Swap	Replaces the face of one individual with another using encoder–decoder GANs.	FaceSwap, DeepFake, DeepFaceLab	• Misinformation • Non-consensual imagery • Impersonation	• Generates realistic alignment and texture blending. • Hard to detect at pixel level.
Attribute Manipulation	Changes face characteristics (age, gender, skin tone & hair style) using conditional GANs or diffusion triggers.	StyleGAN, StarGAN, FaceApp	• Cosmetic enhancement • Fake online profiles	• Subtle local edits that preserve global realism. • Low residual artefacts.
Expression / Lip-Sync Manipulation	Changes expressions or synchronises mouth motions to match the desired audio.	Wav2Lip, Face2Face, Neural Voice Puppetry	• Speech imitation • Fake video calls	• Temporal consistency hides forgery cues • Hard to detect in single frames.
Fully Synthetic Generation	Generates completely original faces that don't resemble actual individuals.	StyleGAN3, Stable Diffusion, DALL·E 2	• Fake personas • Scam profiles • Bot creation	• No original source • High photorealism • Detection solely rely on model artefacts.

7.1.2 Taxonomy of Deepfake Threats

Table 7.2: Taxonomy of Threat Landscape [9][34]

Domain	Examples	Impact	Project Relevance
Individual Privacy	AI-generated dating or scam profiles, identity theft using fake portraits [9].	• Emotional exploitation • Financial fraud • Reputational damage	Drives the need for a user-friendly, privacy-preserving authenticity checker.
Institutional Trust	Business email compromise via deepfaked executives or video calls [13].	• Financial losses • Reputational damage • Low confidence in digital verification.	Justifies the application of explainability and fairness audits to maintain institutional dependability.
Information Integrity	Deepfake propaganda and disinformation on social media [34].	• Public confusion • Low trust in legitimate media evidence	Underlines *FaceGuard*'s greater societal usefulness as a reliable detection and awareness tool.

7.1.3 Analysis of Existing Major Deepfake Datasets

Table 7.3: Comparison Table of Existing Deepfake Datasets [16]

Dataset	Type	Size	Tools / Models	Demographic Diversity	Fairness Metadata	Project Relevance
FaceForensics++ [26]	Video frames	1,000+ videos (approx. 500k frames)	• DeepFake • Face2Face • FaceSwap • NeuralTextures	Limited, western-oriented	No	• Benchmark for traditional CNN models • Lacked demographic breadth & image-only focus.
DeepFake Detection Challenge [7]	Video frames	>100,000 clips (approx. 4M frames)	Various proprietary face-swap pipelines	Moderate	No	• Large-scale dataset • High variation in compression • Video-centric
Celeb-DF [15]	Video frames	5,639 videos (590k frames)	• DeepFake • Low-compression variants	Limited, celebrities-oriented	No	• Higher realism • Synthetic bias toward celebrity features
AI-Face v2	Static images	Approx. 1.65 million	• GANs (StyleGAN, ProGAN) • Diffusion models (Stable Diffusion, DALL-E 2)	Broad gender, skin tone & age group (annotated)	Yes	• Diverse • Fairness-aware • Image-level benchmark
DFD / DFDC Preview	Video frames	363 original and 3,068 fake videos	GANs (DeepFake, FaceSwap)	Limited	No	• Legacy dataset • Not suited for fairness or image-only detection

7.1.4 Analysis of Deepfake Detection Explainability Techniques

Table 7.4: Comparison Table of Deepfake Detection Explainability Techniques [22][18][23]

Criteria	Grad-CAM	LIME	SHAP
Key Principle	Gradient-based class-activation mapping that visualise high-influence pixels contributing to a class score.	Builds a local linear approximation around one prediction to estimate feature contribution.	Uses Shapley-value theory from cooperative game theory to assign quantitative importance to each feature.
Outcome	Heatmap overlay highlighting salient image regions.	Segmented super-pixels with colour-coded contribution weights.	Numerical importance scores and ranked feature list.
Strengths	• Intuitive visualisation • Works seamlessly with CNNs and ViTs • Fast inference	• Model-agnostic • Interprets any classifier • Helps reveal local decision boundaries.	• Theoretically grounded • Provides globally consistent feature attribution.
Limitations	• Broad localisation • Not pixel-precise • Depends on gradient quality	• Computationally slower per prediction • Random perturbations may cause variance.	• High computational cost for large image spaces • Less intuitive for non-experts.
Explainability Level	Visual, region-based interpretability.	Local, per-instance interpretability.	Global, feature-importance interpretability.
Computation Cost	• Low • Real-time	• Moderate • Instance-based	• High • Batch-level
Project Relevance	• Excellent • Fast and GPU-friendly	• Moderate • Usable for on-demand explanations	• Limited • Best for offline audits

7.1.5 Analysis of Deepfake Detection Architectures

Table 7.5: Comparison Table of Deepfake Detection Architectures [14][6][31]

Criteria	Conventional CNNs	Residual / Attention CNNs	Transformer-Based (ViT)	Hybrid CNN–ViT	Graph-Relational & Attention-Enhanced Models
Models	*XceptionNet, MesoNet*	*ResNet, EfficientNet*	*ViT, Swin-T, DeiT*	*ResNet-ViT, EfficientViT*	*GCN-Face, AAS-IST, FAGAN*
Core Mechanism	Learns local pixel-level artefacts (colour mismatch, boundary blur, compression noise).	Residual and squeeze-attention layers enhance deeper feature learning and suppress overfitting.	Self-attention captures long-range spatial dependencies and global illumination triggers.	Fuses CNN's texture sensitivity with ViT's contextual awareness through feature-fusion layers.	Represents facial landmarks as graph nodes with graph-attention or spatial-temporal edges to detect geometric or relational inconsistencies.
Extraction Domain	RGB spatial domain, low–mid-level artefacts.	Deep spatial–frequency domain, hierarchical residual features.	Tokenised patch embeddings in transformer encoder.	Multi-scale hybrid domain (spatial & contextual).	Graph domain (facial landmarks / relational topology).
Training Sets	• Moderate dataset size • Simple GPU training	• Slightly higher complexity • Transfer learning	• Very large and diverse datasets • High computational cost	• Dual-stage training (*CNN* pre-train & *ViT* fine-tune).	• Facial-landmark extraction • Accurate alignment pre-processing
Generalisation & Robustness	• Good within-dataset • Poor cross-dataset generalisation	• Better generalisation • Moderate robustness to compression	• Excellent domain-transfer capability • Stable under generator shifts	• High robustness and fairness stability across datasets	• Robust to spatial distortions and pose changes • Limited by landmark-quality errors
Explainability Integration	Fully compatible with *Grad-CAM*.	Fully compatible with *Grad-CAM*.	Attention-map visualisation and transformer *Grad-CAM* variants.	Compatible with *Grad-CAM* and *SHAP* frameworks.	Explainability through node / edge-attention weights visualisation.
Strengths	• Lightweight and fast with moderate GPU resources. • Highly interpretable convolutional filters reveal texture-level cues (blending boundaries & colour inconsistencies). • Well-supported by	• Skip-connection architecture (*ResNet*) improves gradient flow and supports very deep networks. • Attention modules (*Squeeze-and-Excitation* & *CBAM*) improve focus on salient facial features.	• Self-attention allows modelling of long-range and global relationships across image patches. • Generalises well across unseen generators and domains. • Robust to dataset bias. • Intrinsic attention maps	• Combines *CNN* local-texture precision with *ViT* global-context reasoning for best performance. • Improved robustness to unseen attacks and generative sources. • Balances bias and fairness	• Exploit facial-landmark geometry and topological relationships to detect unrealistic spatial configurations. • Less dependent on pixel texture, resistant to compression artefacts and re-encoding noise.

	PyTorch, *TensorFlow* and pre-trained weights. • Suitable for baseline benchmarking and small-scale experiments.	• Better spatial-frequency representation that captures fine and coarse details simultaneously. • More stable training and reduced overfitting compared with conventional *CNNs*.	naturally support explainability and bias inspection. • Performs well in fairness and demographic-balance evaluations.	metrics. • Supports multiple explainability layers (*Grad-CAM* & *SHAP*) due to dual architecture. • Efficient transfer learning.	• Compact model size and efficient memory usage once landmarks are extracted. • Attention-based graph edges enable interpretable reasoning over facial regions.
Limitations	• Overfits easily to generator-specific artefacts and compression patterns. • Fails to capture global or long-range dependencies between facial regions. • Low accuracy on unseen or cross-dataset manipulations. • Limited capacity to distinguish *DMs* forgeries with globally consistent artefacts.	• High computational load and memory footprint due to deeper stacks and attention layers. • Requires careful hyperparameter tuning. • Limited in capturing non-local contextual dependencies or geometric distortions. • Interpretation of residual activations less intuitive.	• Large samples required for stable convergence. • Very high computational cost and VRAM usage. • Unstable training if without large-scale regularisation or augmentation. • High inference latency than *CNN* counterparts, limiting real-time use.	• Complex architectural with dual-branch fusion • Large training requires for large GPU resources and careful layer-freeze scheduling. • Longer training time • High risk of instability in early epochs. • Varying model interpretability depending on fusion-layer design.	• Reliant on accurate facial-landmark detection • Degraded performance under occlusion or pose variation. • Pre-processing pipeline adds latency (landmark extraction & graph construction). • Limited availability of annotated landmark-graph datasets. • Scaling to high-resolution imagery or multi-face scenes is computationally challenging.
Project Relevance	Baseline detector for benchmarking.	Adopted as core detection backbone.	Acts as global-context encoder.	Integrated for bias mitigation.	Considered for future extension to geometric anomaly detection.

7.1.6 Analysis of Project Process Model

Table 7.6: Comparison Table of Project Management Model

Process Model	Key Characteristics	Advantages	Disadvantages	Project Relevance
Waterfall Model [27]	Linear sequential flow (requirements, design, implementation, testing, maintenance).	• Clear structure and strong documentation • Good for fixed-scope projects with stable requirements	• Rigid • Hard to adapt to changes • High cost of error correction • Impractical for experimental AI workflows	• Low • Lacked flexibility for iterative AI model testing and fairness refinement
Spiral Model [2]	Iterative prototyping with risk analysis at each phase.	• Incorporates risk management and stakeholder review • Flexible for research-driven projects	• Heavy documentation • High planning overhead • Complex for small academic teams	• Moderate • Conceptually strong • Impractical for university-level project and resource constraints
Agile Scrum [28]	Incremental, sprint-based development with constant feedback and continuous expansion.	• Highly adaptive • Supports early prototyping • Allows stakeholder collaboration and rapid iteration	• Disciplined communication • Backlog maintenance.	• High • Best balance of flexibility with evolving AI goals

7.1.7 Gap Analysis of Existing Deepfake Models and *FaceGuard* Design

Table 7.7: Comparison Table of Deepfake Detection Explainability Techniques

Aspect / Criterion	Existing Models / Literature Reviews	*FaceGuard* Design
Training Dataset	Earlier benchmarks (*FF++, DFDC, Celeb-DF*) focus on video frames and older GANs, limited coverage of diffusion-model fakes.	***AI-Face*** dataset with manipulated images from *GANs* and *DMs* for broader generalisation and fairness benchmarking.
User Interface	Existing academic projects lack usable, explainable web interfaces.	***React*-based dashboard** with intuitive upload and confidence score.
Robustness	CNN-based detectors overfit to dataset-specific artefacts and degrade on unseen attacks or compression noise.	**MVP 1: RestNet-based model** for baseline model. **MVP 2: Hybrid CNN + Vision Transformer** for spatial-frequency robustness and domain generalisation.
Fairness and Demographic Bias	Detection models often show unequal accuracy across gender and skin tone due to biased datasets and lack of subgroup auditing.	*AI-Face* dataset with **demographic annotation** to perform fairness evaluation and threshold calibration for balanced accuracy by subgroup.
User Transparency	"Black box" detectors with binary outcomes without rationale.	**Grad-CAM** saliency visualisation and confidence scoring.
Privacy and Data Protection	Online tools frequently upload and store user images, non-compliance with GDPR and PDPA.	**Ephemeral, in-memory processing** with automatic deletion, stripped EXIF metadata, HTTPS encryption and secure authentication.
Project Management	Rigid development cycles that cannot adapt to evolving AI datasets or ethics.	**Agile Scrum model** with five sprints, enabling iterative refinement of model training, UI/UX and performance.

7.2 Requirements Traceability Matrix (RTM)

Table 7.8: Requirement Traceability Matrix (RTM)

REQ	Description	Categories	FR / NFR	Source	Priority	Status
R01	The training model should classify uploaded facial images as real or AI-generated so that users can verify the authenticity of online profile photos.	Model Training	FR	Project Specification, Supervisor	**High**	Pending
R02	The training model should be trained using large-scale datasets containing real and AI-generated faces to improve detection accuracy and generalisation.	Model Training	FR	Project Specification, Supervisor	**High**	Pending
R03	The web application should allow users to securely upload face images for analysis through a drag-and-drop interface.	Web Application	FR	Project Specification, Supervisor	**High**	Pending
R04	The web application should validate uploaded files to ensure only supported formats (JPEG, PNG, WEBP) are accepted to prevent invalid or unsafe uploads.	Security Concern	FR	Supervisor	**High**	Pending
R05	The web application should automatically remove uploaded images from temporary storage after processing to protect user privacy.	Security Concern	NFR	Project Specification, ACM Code of Ethics	**High**	Pending
R06	The application should provide a confidence score alongside each classification result so that users can interpret the model's certainty.	Web Application	FR	Project Specification, Supervisor	**High**	Pending
R07	The application should display explainability visuals, such as saliency maps, to help users understand which regions influenced the model's decision.	Web Application	FR	Project Specification, Supervisor	**High**	Pending
R08	The web interface should present classification results, confidence scores and explanations in a clean, organised and visually accessible dashboard.	Usability & User Experience	FR	Project Specification, Supervisor	**High**	Pending
R09	The web application should include authentication and login features that allow registered users to log in securely and access the web smoothly.	Usability & User Experience	NFR	Supervisor	**Moderate**	Pending
R10	The web application should display clear error messages when users attempt to upload invalid or unsupported image files.	Usability & User Experience	NFR	Supervisor	**Moderate**	Pending
R11	The web interface should include a progress indicator or loading animation to inform users that the uploaded image is being processed at what stage.	Usability & User Experience	NFR	Supervisor	**Moderate**	Pending

R12	The system should display an appropriate message if the model or server is temporarily unavailable to ensure clear communication to users.	Usability & User Experience	NFR	Supervisor	**Moderate**	Pending
R13	The web application should enforce upload file size limits to maintain system performance and prevent excessive resource usage.	Performance & Scalability	NFR	Supervisor	**Moderate**	Pending
R14	The system should maintain stable inference speed to allow users to receive classification results without noticeable delay.	Performance & Scalability	NFR	Supervisor	**Moderate**	Pending
R15	The system should be capable of handling multiple concurrent user requests without degradation in performance.	Performance & Scalability	NFR	Supervisor	**Moderate**	Pending
R16	The system should reject malicious or adversarial image uploads to prevent model exploitation or security breaches.	Security Concern	NFR	Project Specification, ACM Code of Ethics	**Moderate**	Pending
R17	The system should support HTTPS encryption for secure data transmission between the client and server.	Security Concern	FR	Supervisor	**Moderate**	Pending
R18	The web application should include disclaimers reminding users that *FaceGuard*'s results are advisory and should not be used for personal identification.	Security Concern	NFR	Project Specification, ACM Code of Ethics	**Moderate**	Pending
R19	The training model should evaluate performance across demographic groups to ensure fairness and inclusivity in its predictions.	Model Evaluation	FR	Project Specification, Supervisor	**Low**	Pending
R20	The training model should apply bias mitigation techniques during training or inference so that subgroup performance remains balanced and equitable.	Model Training	FR	Project Specification, Supervisor	**Low**	Pending

7.3 Product Acceptance Criteria

Table 7.9: Product Acceptance Criteria

ID	User Story	User Acceptance Criteria (UAC)
US1	As a user, I want to upload a profile picture easily so that I can verify whether it is real or AI-generated.	The upload feature accepts only image formats (.jpg, .png, .webp). Upload must be completed in ≤ 2 clicks (drag/drop or browse). Invalid or corrupted files trigger an error message.
US2	As a user, I want to receive an authenticity classification result so that I can identify fake or AI-generated faces before trusting a profile.	The AI model classifies images as "Real" or "AI-generated" with ≥ 90% accuracy. Results appear on the dashboard within 2-3 seconds of upload. Each result displays a confidence score (0–1 scale).
US3	As a privacy conscious user, I want to ensure my uploaded images are not stored or shared so that my personal data remains protected.	All uploads are processed in volatile storage and deleted post-inference. HTTPS encryption is enforced for all data transfers. No user data is retained in logs or databases.
US4	As a non-technical user, I want to see an explanation on why the image was classified as fake or real so that I can trust and understand the AI's reasoning.	Grad-CAM or equivalent explainability visualisation displayed for predictions. Heatmaps clearly highlight regions influencing classification. Legends or tooltips describe what the visual represents.
US5	As a user, I want to view the result in a clean and easy to understand dashboard so that I can interpret the output without technical knowledge.	Dashboard layout displays image, result, confidence, and heatmap in a structured format. Responsive interface adapts to major browsers and screen sizes. Interface adheres to accessibility standards (ie, text contrast, labelled visuals).
US6	As a registered user, I want to log in securely so that I can access additional features and protect my analysis history.	Login and registration validated via secure authentication (OAuth2). Invalid credentials trigger a proper error message. Session tokens automatically expire after 30 minutes of inactivity.
US7	As a supervisor or stakeholder, I want to verify that the system behaves fairly across demographics so that its deployment complies with research and ethical standards.	Bias-mitigation techniques applied during training. Model tested on diverse datasets with ≤ 5% accuracy variance across subgroups. The system includes a visible disclaimer that results are advisory.
US8	As a user, I want to be informed about the system status during processing so that I know when my image is being analysed or if an error occurs.	Progress indicator displayed during image analysis. Messages shown for "Processing", "Completed", and "Error". The system recovers if the backend is temporarily unavailable.

US9	As a user, I want to know that the website is secure and reliable so that I can trust the system for sensitive uploads.	HTTPS is implemented site-wide. Backend hosted on a verified platform with uptime ≥ 99%. Security testing performed and documented during the testing phase.
US10	As a supervisor, I want to evaluate system performance and functionality so that I can confirm final sign-off.	All functional and non-functional requirements in the RTM are met. No critical bugs remain open in Jira. Feedback is ≥ 85% in user satisfaction during evaluations.

7.4 RACI Matrix

Table 7.10: RACI Table

Key Task	Product Owner	Technical Specialist	Backend Specialist	QA/QC Specialist	Supervisor
Project planning	A	R	I	I	C
Dataset preprocessing	I	A	C	R	C
Model development	I	R	C	I	C
Backend API integration	I	A	R	C	C
User interface development	I	C	I	R	C
Model testing	I	C	I	R	C
System integration testing	I	C	R	A	C
User acceptance testing	R	C	I	A	C
Final documentation	A	R	C	C	C
Stakeholder communication	R	I	I	I	A

***RACI Matrix - R: Responsible, A: Accountable, C: Consulted and I: Informed.

7.5 Team Organisation Structure

Table 7.11: MCS24 Team Organisation Structure

Our Team	Roles and Responsibilities
 Er Jun Yet *Product Owner & Team Lead*	• Articulates the overall product vision and goals for the *FaceGuard* project. • Works closely with the development team to clarify requirements and lead sprint progress. • Monitors overall project progress, engages stakeholders and ensures timely completion of deliverables. • Creates, maintains and prioritises the product backlog according to project value and stakeholder interests. • Establish clear acceptance criteria for all user stories to ensure all deliverables meet project objectives and quality standards.
 Tan Yan Yi *Scrum Master & Technical Lead*	• Hosts and facilitates Agile sprint ceremonies including sprint planning, stand-ups, sprint review and sprint retrospective. • Guide and support the development team in Scrum practices. • Address development blockers that hinder the team progress. • Provide guidance and oversight on all technical aspects of the *FaceGuard* project, including architectural design. • Establish technical documentation and enforce coding standards in development process.

 Bryan Ong Zong Sheng *Backend Specialist &* *Machine Learning Specialist*	• Designs, develops and maintains the backend infrastructure using *FastAPI* for secure data processing and model deployment. • Builds, trains and fine-tunes deep learning models (*ResNet* and Hybrid *CNN–ViT*) for deepfake image detection. • Integrates the trained AI model with the web application to enable real-time, privacy-preserving inference. • Monitors system performance, evaluates model accuracy and fairness metrics and continuously improves backend and AI components.
 Muhammad Ali Afzal *UI/UX Specialist &* *QA/QC Specialist*	• Designs and develops an intuitive React.js, user-friendly web interface that aligns with *FaceGuard*'s theme and also *Grad-CAM* visualisation. • Creates mockup interface to visualise user flows and enhance usability. • Conducts user experience testing to ensure usability and responsiveness of the FaceGuard platform prior end of sprint. • Develops and executes quality assurance test plans to identify and resolve functional or visual defects.

7.6 Communication Management Plan

Table 7.12: Communication Register

Communication Type	Purpose	Platform	Frequency	Participants	Chairperson	Deliverable	Output Format
Standup Meeting	Update on project progress and inform upcoming tasks	Zoom / WhatsApp	Weekly	MCS24 Team	Scrum Master	Document the action steps	Google Drive
Sprint Planning	Plan tasks for the next sprint and delegate tasks	Zoom	Start of sprint	MCS24 Team, Supervisor	Scrum Master	Update the sprint backlog and task delegation	Jira
Sprint Retrospective	Inspect the team process and make plans for improvement	Zoom	End of sprint	MCS24 Team	Scrum Master	Note the lessons learned	Google Drive
Sprint Review	Inspect the product increment and adapt based on supervisor feedback	Zoom	End of sprint	MCS24 Team, Supervisor	Product Owner	Note the feedback given	Google Drive
Emergency Meeting	Solve any blockers hindering progress	Zoom / WhatsApp	Depends	MCS24 Team	Project Owner	No deliverable	WhatsApp

7.7 Code Review Checklist

Table 7.13: Code Review Checklist

Category	Item	Tick
Code Quality & Standards	Code adheres to the project's naming conventions and formatting requirements	**[ ] Pass [ ] Fail**
	No unused imports, variables or redundant code.	**[ ] Pass [ ] Fail**
	Functions and methods are concise, modular and single-purpose.	**[ ] Pass [ ] Fail**
	Code includes relevant comments and docstrings that explain difficult reasoning.	**[ ] Pass [ ] Fail**
	All constants, API keys and environment variables are extendable and not hard-coded.	**[ ] Pass [ ] Fail**
Functionality & Testing	Code successfully runs all covered unit tests without issues.	**[ ] Pass [ ] Fail**
	Edge cases and invalid input scenarios are properly handled.	**[ ] Pass [ ] Fail**
	Integration tests ensure correct data flow between backend, AI model and frontend.	**[ ] Pass [ ] Fail**
	The model inference endpoint correctly outputs accurate and desired results.	**[ ] Pass [ ] Fail**
Security & Privacy	No sensitive data (user uploads or API tokens) is tracked or stored permanently.	**[ ] Pass [ ] Fail**
	Input validation and sanitisation are implemented to prevent injection attacks.	**[ ] Pass [ ] Fail**
	HTTPS or relevant encryption utilised for all data transmissions.	**[ ] Pass [ ] Fail**
	Uploaded images are processed in-memory and automatically deleted after inference.	**[ ] Pass [ ] Fail**
Performance & Efficiency	Code executes efficiently without unnecessary loops or computations.	**[ ] Pass [ ] Fail**
	Model inference latency within the acceptable range.	**[ ] Pass [ ] Fail**
	No excessive console logs or debug prints in production code.	**[ ] Pass [ ] Fail**

Ethical & Explainability Compliance	Grad-CAM heatmaps correctly generate visual outputs for explainability.	**[ ] Pass [ ] Fail**
	Model behaviour aligns with fairness and bias-mitigation goals.	**[ ] Pass [ ] Fail**
	All code changes are reviewed and approved by QA/QC and Product Owner before merging.	**[ ] Pass [ ] Fail**

7.8 Risk Register

Table 7.14: Risk Register

Description	Date Raised	Impact	Likelihood	Severity	PIC	Monitoring Strategy	Mitigation Strategy	Contingency Plan
Project Risk								
Academic schedules affecting the progress of the project and sprint deadlines	31/10/2025	**High**	**High**	**High**	Team Lead	Regular standup meetings and sprint planning to monitor the progress	Use agile methodology to reallocate the tasks and adjust the sprint backlog	Have buffer time for the project timeline where non-critical features can be delayed if needed
Project scope changes from supervisor requests	31/10/2025	**High**	**Medium**	**Medium**	Team Lead	Have a strict control process change using the Product Backlog	Get the sign-off on a clear scope document during the project inception	Discuss the timeline and resources and prioritise the backlog from the stakeholder feedback
Technical Risk								
Model fails to meet the target benchmark accuracy	31/10/2025	**High**	**High**	**High**	Technical Lead	Track the accuracy validation at the end of each sprint	Experiment with different model architectures and learning progress early.	Change the benchmark accuracy to be lower
Integration of AI model and web API fails	31/10/2025	**High**	**Medium**	**Medium**	Backend Specialist	Test the integration starting from the first sprint	Use Docker to ensure it is reliable across different environments and define the API early.	Implement a fallback for demonstration or use a different API.

Unable to find enough AI-face generated images for training	31/10/2025	**High**	**High**	**High**	QA Specialist	Monitor the diversity of the datasets every two weeks	Use different types of data sources such as academic datasets	Use data augmentation techniques to have value of the current datasets
Model has bias against certain groups	31/10/2025	**High**	**Medium**	**Medium**	QA Specialist	Implement bias detection and testing on those groups	Apply bias mitigation techniques while training the model	Retrain the model using a balanced dataset or properly document the limitations of the model
Computational cost of the training exceeds the expected budget	31/10/2025	**Medium**	**Medium**	**Medium**	Team Lead	Monitor the network usage of the services weekly	Optimise the code to be more efficient or create burner accounts for more network limits.	Use local machine to train or apply for educational credits
Operational Risk								
Certain team members become unavailable due to emergency	31/10/2025	**High**	**Medium**	**Medium**	Team Lead	Track the members availability using Jira	Contain clear documentation on the critical components	Distribute the tasks left to the remaining members and inform the supervisor
Team has conflict which affects productivity	31/10/2025	**Medium**	**Low**	**Low**	Team Lead	Observe the team collaboration during meetings	Encourage an open feedback culture to prevent misunderstandings	Inform the team leader to fix the issue and inform the supervisor if it continues
Learning curve for the new tools used	31/10/2025	**Medium**	**Medium**	**Medium**	Technical Lead	Track the progress using developer spikes	Give learning time during the sprint and share knowledge with other members on the tools	Find courses or resources online and ask supervisor for help

7.9 SWOT Analysis

Table 7.15: SWOT Analysis

Strength	The whole team is quite interested in the project which would make discussions and collaboration much smoother.
Weakness	Only one member has deep learning knowledge in training the model while the other members has little to no experience which could hinder the progress.
Opportunities	Deepfake detection emerging open-source models could be a potential for publication.
Threats	Price changes in Google Colab or network limit reached in GPU access would prevent progress.

7.10 TARA Framework

Table 7.16: TARA Framework

Transfer	Shifting the impact or the responsibility of the risk to a third party like using cloud services such as Google Colab in case of hardware failure.
Avoid	Eliminating the risk by not doing the task that caused the risk like using an unstable library and using well-known ones like PyTorch.
Reduce	Implementing measures to reduce the chances of the risk happening like implementing the logic early and doing frequent tests to reduce the potential bugs.
Accept	Acknowledging the risk and preparing for the impact of the risk like when the cloud services go into maintenance and waiting is inevitable.

7.11 Work Breakdown Structure (WBS)

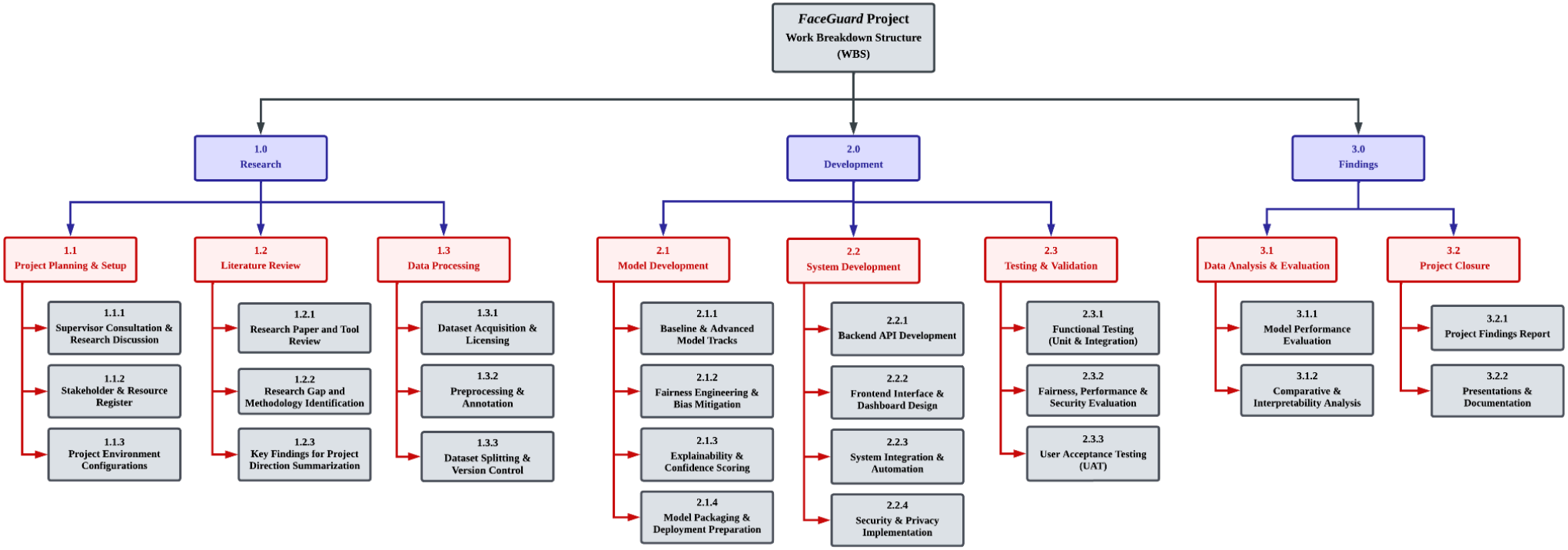

Figure 7.1: Work Breakdown Structure (WBS)

7.12 Gantt Chart

Name	Begin date	End date
Gantt Chart	18/8/25	28/5/26
Research Phase	18/8/25	29/10/25
Project Planning & Setup	18/8/25	26/8/25
Supervisor Consultation and Research Discussion	18/8/25	25/8/25
Stakeholder and Resource Register	26/8/25	26/8/25
Literature Review	27/8/25	26/9/25
Research Paper and Tool Review	27/8/25	5/9/25
Research Gap and Methodology Identification	8/9/25	16/9/25
Key Findings for Project Direction Summarization	17/9/25	26/9/25
Data Processing	29/9/25	29/10/25
Dataset Acquisition & Licensing	29/9/25	2/10/25
Preprocessing & Annotation	3/10/25	28/10/25
Dataset Splitting & Version Control	29/10/25	29/10/25
Dataset Finalization	30/10/25	30/10/25

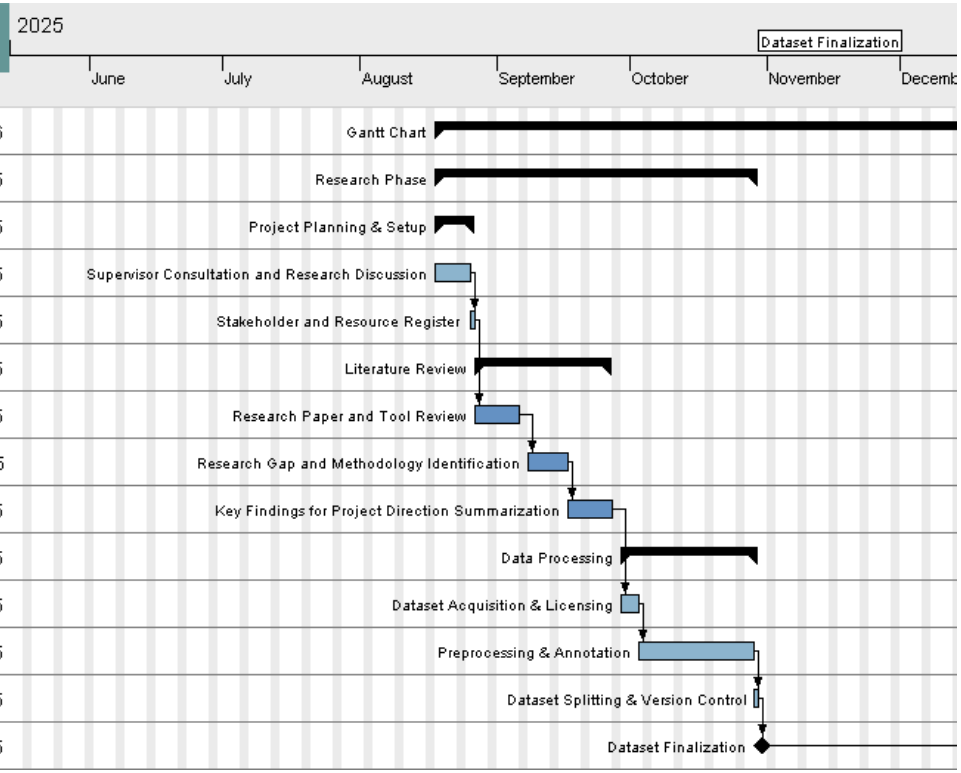

Figure 7.2: Gantt Chart - Expending for Research

Name	Begin date	End date
Development Phase	2/3/26	19/5/26
Sprint Cycle	2/3/26	19/5/26
Sprint Planning	2/3/26	2/3/26
Sprint Execution	2/3/26	19/5/26
Model Development	2/3/26	27/3/26
Baseline & Advanced Model Tracks	2/3/26	6/3/26
Fairness Engineering & Bias Mitigation	9/3/26	13/3/26
Explainability & Confidence Scoring	16/3/26	20/3/26
Model Packaging & Deployment Preparation	23/3/26	27/3/26
System Development	30/3/26	6/5/26
Backend API Development	30/3/26	10/4/26
Frontend Interface & Dashboard Design	13/4/26	20/4/26
System Integration & Automation	21/4/26	28/4/26
Security & Privacy Implementation	29/4/26	6/5/26
Testing & Validation	7/5/26	19/5/26
Functional Testing (Unit & Integration)	7/5/26	8/5/26
Fairness, Performance & Security Evaluation	11/5/26	15/5/26
User Acceptance Testing (UAT)	18/5/26	19/5/26
Product Deployment	20/5/26	20/5/26
Finding Phase	20/5/26	28/5/26
Data Analysis & Evaluation	20/5/26	22/5/26
Model Performance Evaluation	20/5/26	20/5/26
Comparative & Interpretability Analysis	21/5/26	22/5/26
Project Closure	25/5/26	28/5/26
Prepare Findings Report	25/5/26	25/5/26
Presentations & Documentation	26/5/26	28/5/26
Final Review & Reflection	29/5/26	29/5/26

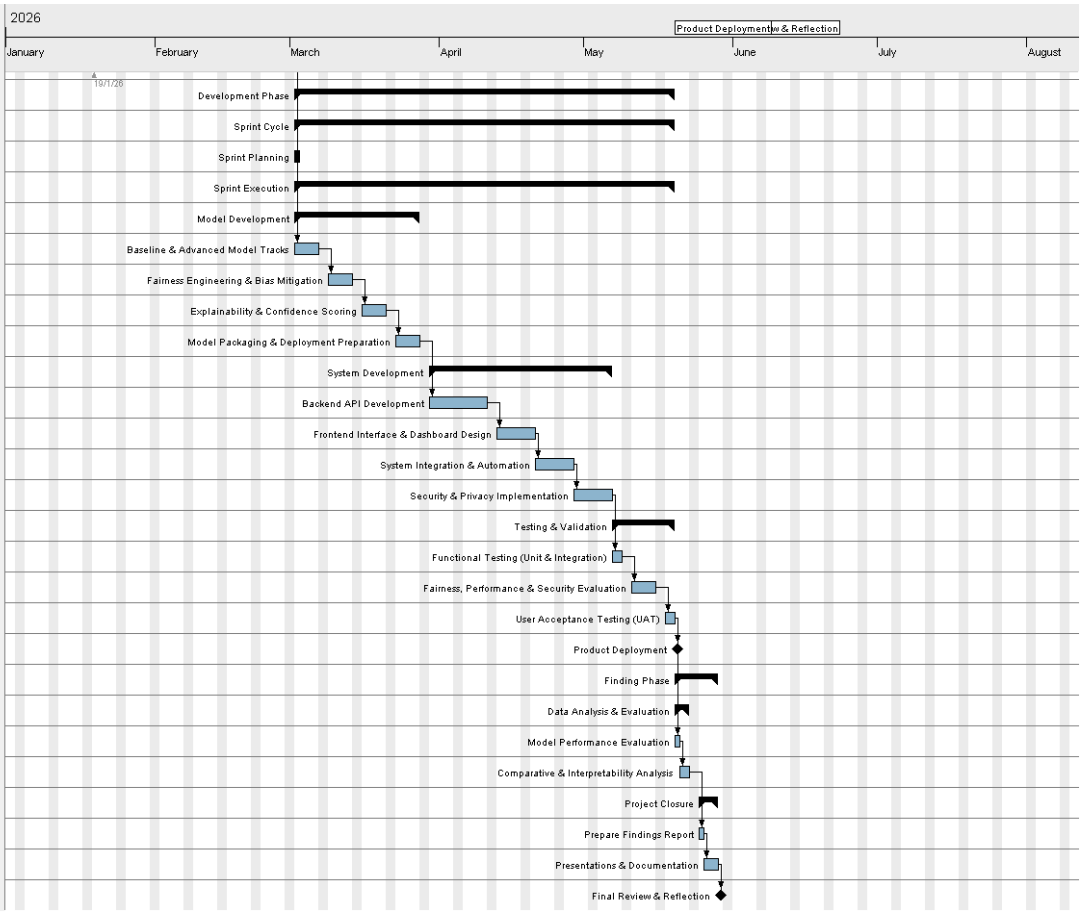

Figure 7.3: Gantt Chart - Expanding for Development and Findings

7.13 Commitment Register

Table 7.17: Commitment Register

Sprint Goals	Expected Hours / Week	Duration	Total Hours / Member
Research & Requirement Analysis	3 h	Week 1	3 h
Model Development & Testing	6 h	Weeks 2 - 6	30 h
Web Integration & API Development	5 h	Weeks 7 - 10	20 h
UI Design & Usability Testing	4 h	Weeks 11 - 12	8 h
Sprint Review & Documentation	3 h	Throughout Project	15 h
Total Estimated Workload / Member	≈ 21 hours per week	12 weeks	≈ 252 hours

End of Proposal